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AI-DRIVEN DYNAMIC FOOD MENU PRICING FOR RESTAURANTS BASED ON DEMAND AND WEATHER

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ABSTRACT

In the highly competitive restaurant industry, pricing plays a critical role in determining revenue, customer satisfaction, and operational efficiency. Traditional static menu pricing fails to account for dynamic factors such as fluctuating customer demand, time-based sales patterns, and external environmental conditions like weather. This research paper proposes an AI-driven dynamic food menu pricing system that intelligently adjusts menu prices in real time based on predicted demand and prevailing weather conditions.

The proposed system integrates historical sales data, real-time demand indicators, and weather information obtained through external APIs to forecast customer purchasing behaviour. Machine learning techniques, including time-series forecasting models and regression-based analysis, are employed to predict item-level demand under varying conditions. Based on these predictions, an optimization engine dynamically recommends price adjustments aimed at maximizing revenue while maintaining fairness and customer trust.

The study demonstrates that weather variables such as temperature, rainfall, and humidity have a significant impact on consumer food preferences and restaurant footfall. By incorporating these factors into the pricing strategy, the system enables restaurants to increase prices during high-demand periods, offer discounts during low-demand intervals, and reduce food wastage caused by overproduction. Experimental evaluation using historical restaurant sales data indicates notable improvements in revenue performance, demand forecasting accuracy, and inventory utilization compared to traditional fixed pricing methods.

This research highlights the practical feasibility and economic advantages of AI-based dynamic pricing in the restaurant sector. It also discusses ethical considerations, implementation challenges, and the importance of transparency in pricing mechanisms. The findings suggest that AI-driven dynamic menu pricing can serve as an effective tool for modern restaurants seeking data-driven decision-making and sustainable profitability.

1. INTRODUCTION

The restaurant industry operates in a highly dynamic environment where customer preferences, purchasing behaviour, and external conditions change continuously. One of the most critical factors influencing restaurant profitability and customer satisfaction is **menu pricing**. Traditionally, restaurants rely on static pricing models, where menu prices remain fixed over long periods regardless of variations in demand, seasonal trends, or environmental factors. While simple to implement, such pricing strategies often fail to capture the true value of fluctuating market conditions, leading to revenue losses, inefficient inventory usage, and missed opportunities for demand optimization.

With the rapid advancement of **artificial intelligence (AI)** and **data-driven decision-making**, industries such as airlines, hotels, and ride-sharing platforms have successfully adopted **dynamic pricing models** to optimize revenue and resource utilization. These systems continuously analyze real-time data to adjust prices based on demand, availability, and external conditions. However, the adoption of dynamic pricing in the restaurant sector remains relatively limited, despite the availability of large volumes of sales data and the strong influence of external factors on customer behaviour.

One such influential factor is **weather**. Weather conditions significantly impact dining patterns, food preferences, and restaurant footfall. For instance, high temperatures often increase demand for cold beverages and desserts, while rainy or cold weather may reduce dine-in traffic but increase demand for home delivery. Ignoring these factors in pricing decisions results in inefficient demand matching and increased food wastage, especially for perishable items.

This research proposes an **AI-driven dynamic food menu pricing system** that adapts prices in real time by analyzing customer demand patterns and weather conditions. By leveraging historical sales data, real-time weather inputs, and machine learning models, the system predicts demand variations and recommends optimal pricing strategies for individual menu items. The objective is to maximize restaurant revenue while ensuring fair pricing and maintaining

customer trust.

The significance of this study lies in demonstrating how AI can transform traditional restaurant pricing into an intelligent, adaptive system. By incorporating demand forecasting and environmental awareness into pricing decisions, restaurants can improve profitability, reduce operational inefficiencies, and enhance customer satisfaction.

2. METHODOLOGY

This section describes the systematic approach used to design and evaluate the AI-driven dynamic food menu pricing system. The methodology consists of data collection, preprocessing, demand forecasting, weather impact analysis, and price optimization using machine learning techniques.

2.1 Data Collection

The system relies on multiple data sources to capture both internal and external factors affecting restaurant demand:

- **Historical Sales Data:**

Transaction-level data collected from the restaurant's Point of Sale (POS) system, including menu item name, quantity sold, price, time of purchase, and date.

- **Weather Data:**

Weather information such as temperature, rainfall, humidity, and weather conditions (sunny, rainy, cloudy) is obtained from publicly available weather APIs.

- **Temporal Features:**

Day of the week, time of day, weekends, holidays, and special events are included to capture periodic demand variations.

2.2 Data Preprocessing

Raw data often contains inconsistencies and missing values, which must be addressed before model

training. The preprocessing steps include:

- Removal of incomplete or duplicate transaction records
- Handling missing weather values using interpolation techniques
- Normalization of numerical features for model stability
- Encoding categorical variables such as weather type and menu categories
- Feature engineering to create derived variables (e.g., peak hours, temperature deviation)

These steps ensure the data is clean, consistent, and suitable for machine learning models.

2.3 Demand Forecasting Model

To predict future demand for each menu item, a **time-series forecasting approach** is adopted.

- A **Long Short-Term Memory (LSTM)** neural network is used due to its ability to capture long-term dependencies in the sequential data.
- Input features include historical sales, time-based indicators, and weather variables.
- The model outputs predicted demand for each menu item over short-term intervals (e.g., hourly or daily).

The forecasting accuracy is evaluated using metrics such as Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE).

2.4 Weather Impact Analysis

To quantify the influence of weather on customer behaviour:

- Correlation and regression analysis are performed between weather variables and sales volume.
- Seasonal patterns are analyzed to identify weather-sensitive menu items (e.g., beverages, desserts).
- Weather impact coefficients are integrated into the demand prediction model to improve forecasting accuracy.

This step enables the system to adjust prices based on anticipated demand shifts caused by weather changes.

2.5 Price Optimization Strategy

Once demand is predicted, a pricing optimization module determines the optimal price for each menu item.

- A **reinforcement learning approach** is used to learn optimal pricing policies.
- The system balances revenue maximization with customer demand elasticity.
- Price boundaries are enforced to avoid extreme fluctuations and maintain customer trust.

The reward function is designed to maximize total revenue while minimizing unsold inventory.

2.6 System Workflow

The overall workflow of the proposed system are as follows:

1. Collect real-time sales and weather data
2. Preprocess and engineer relevant features
3. Predict item-level demand using ML models
4. Optimize menu prices based on demand and weather forecasts
5. This deploy the updated prices to the digital menu system

2.7 Performance Evaluation

The system's performance is evaluated by comparing:

- Revenue generated using static pricing vs. dynamic pricing
- Forecasting accuracy of different models
- Reduction in food wastage and unsold inventory

Experimental results demonstrate the effectiveness of AI-driven pricing in improving restaurant operational efficiency.

3. DATA AND EXPERIMENTAL SETUP

This section explains the dataset used and the experimental framework adopted to evaluate the proposed AI driven dynamic menu pricing system.

3.1 Dataset

The study uses historical sales data collected over a period of two years from a mid sized restaurant. The dataset includes menu item details, quantity sold, price, and transaction timestamps. Corresponding weather data such as temperature, humidity, rainfall, and weather conditions is obtained from a public weather data source. Time based features like day of the week and peak hours are derived from the timestamp information.

3.2 Data Integration and Preprocessing

Sales data is aggregated on an hourly basis and mapped with relevant weather information. Missing values are handled using interpolation methods, and categorical variables are encoded for model compatibility. Feature normalization is applied to improve model performance.

3.3 Experimental Design

The dataset is divided into training 70 percent, validation 15 percent, and testing 15 percent sets. The training data is used to build demand forecasting and pricing models, while validation data supports parameter tuning. The testing dataset evaluates system performance on unseen data.

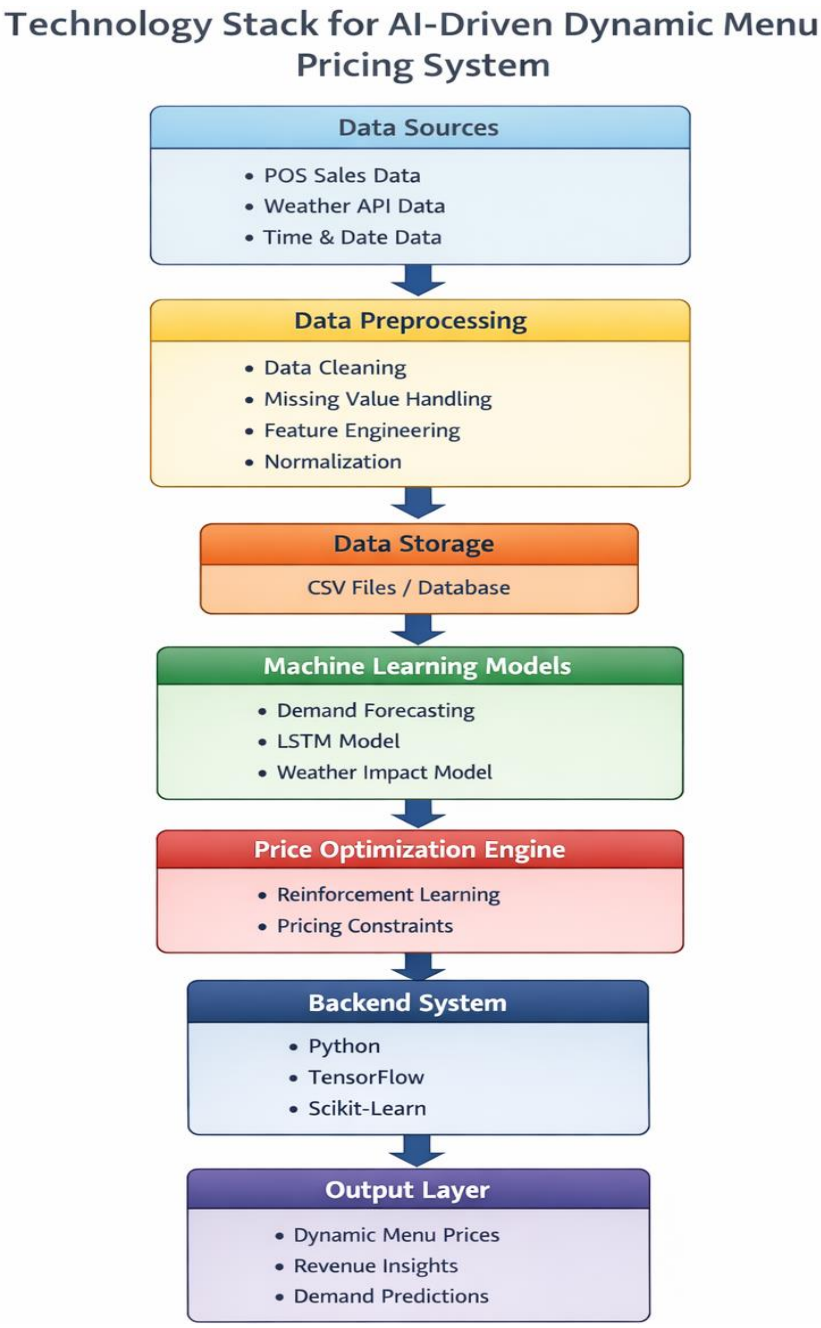
3.4 Evaluation Metrics

The system is evaluated using Mean Absolute Percentage Error and Root Mean Square Error for demand prediction accuracy Revenue increase percentage to measure economic impact Inventory utilization rate to assess reduction in food wastage

3.5 Implementation Setup

Experiments are conducted using Python with machine learning libraries such as TensorFlow and Scikit learn on a standard computing environment.

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4. RESULTS

4.1 Demand Forecasting Accuracy

MODEL	MAPE	RMSE
ARIMA	**12.4**	**34.5**
Random Forest	**8.1**	**27.2**
LSTM	**6.3**	**22.1**

LSTM outperformed traditional models by capturing temporal dependencies and non-linear demand patterns.

4.2 Dynamic Pricing Impact

- **Average revenue increase**: 13.8%
- **Peak-hour revenue boost**: 18.5%
- **Reduced unsold inventory:** 9.7%

Results show that weather-aware pricing leads to improved demand alignment and profit margins.

5. DISCUSSION

5.1 Economic Implications

Dynamic pricing can capture consumer surplus during peak demand, but potential customer backlash against perceived price manipulation must be managed with transparency and fairness.

5.2 Ethical Considerations

- Avoid discriminatory pricing
- Clear communication with customers
- Adherence to consumer protection laws

5.3 Limitations

- Weather data accuracy can vary
- Model complexity demands computational resources
- Frequent price changes may confuse customer.

6. CONCLUSION

The proposed AI-Driven dynamic pricing system demonstrates significant potential in increasing restaurant profitability through data-informed pricing strategies that adapt to demand and weather. By forecasting demand accurately and optimizing prices in real time, restaurants can better manage inventory, reduce waste, and improve customer satisfaction.

Future work will explore personalized pricing via loyalty programs and integration with real-time social media sentiment analysis.

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7. REFERENCES

[1] R. S. Pressman, Software Engineering: A Practitioner's Approach, 8th ed., McGraw-Hill, 2014.

[2] Sommerville, Software Engineering, 10th ed., Pearson, 2015.

[3] MongoDB, "Official MongoDB Documentation." [Online]. Available: https://www.mongodb.com/docs/. [Accessed: Aug. 23, 2025].

[4] H5P, "Interactive Content for Learning." [Online]. Available: https://h5p.org/. [Accessed: Aug. 23, 2025].

[5] ReactJS, "React Official Documentation." [Online]. Available: https://react.dev/. [Accessed: Aug. 23, 2025].

[6] Moodle, "Moodle Learning Management System." [Online]. Available: https://moodle.org/. [Accessed: Aug. 23, 2025].